

Ramzan Basheer

Machine Learning Researcher | Deep Learning, Graph Learning, Time-Series

Thiruvananthapuram, India | r4ramzan@gmail.com | +91-9769171579

linkedin.com/in/r4ramzan-basheer | github.com/ramzan-basheer

ramzan-basheer.github.io | scholar.google.com/citations?user=qPyFnIAAAAAJ

orcid.org/0000-0001-5856-5141

PhD expected Dec 2026 | Available Jan 2027



PROFESSIONAL SUMMARY

Machine learning researcher developing deep learning architectures for structured, relational, and temporal data. Research spans graph and hypergraph neural networks, geometric deep learning, state-space models, equivariant learning, and manifold-based methods, with a focus on structural inductive biases. Published 6 peer-reviewed papers (5 first-author), including Hypercube-S4GNN for strong performance in seizure detection on CHB-MIT and TUH. Strong background in PyTorch/PyG and reproducible ML research. PhD candidate at IIST; M.Tech., IIT Bombay; B.Tech., NIT Calicut; prior software engineering at Oracle and Infosys.

TECHNICAL SKILLS

Machine Learning: Machine Learning, Deep Learning, Representation Learning, Graph Machine Learning, Geometric Deep Learning, Continual Learning, Equivariant Learning, Riemannian/Manifold Learning

Deep Learning Architectures: Graph Neural Networks, Hypergraph Neural Networks, State-Space Models (S4, Mamba), Attention Mechanisms, Kolmogorov–Arnold Networks

Modern AI: Transformers, LLMs, fine-tuning

Programming: Python, C/C++, Java, MATLAB

ML Frameworks: PyTorch, PyTorch Geometric, TensorFlow, DGL, scikit-learn, Hugging Face

Scientific Computing: NumPy, SciPy, pandas, MNE-Python, EEGLAB, pyEDFlib

Development Tools: Git, Docker, Linux, Weights & Biases, PyTorch Lightning, model serving

RESEARCH & R&D EXPERIENCE

Senior Research Fellow

Aug 2024 – Present

Junior Research Fellow

Aug 2022 – Jul 2024

CVVR Lab, Indian Institute of Space Science and Technology (IIST), Thiruvananthapuram, India

- **Geometric and multi-edge graph modeling:** Designed Hypercube-S4GNN, a multi-edge graph neural architecture that combines relational graph structure with state-space sequence modeling to learn joint spatial and temporal representations from multivariate signals. Achieved state-of-the-art seizure detection performance on CHB-MIT and TUH (*IEEE MLSP*, 2024). Implemented in PyTorch Geometric.
- **Higher-order representation learning:** Developed a neural state-space hypergraph network to model higher-order interactions and distributional structure beyond pairwise graph relationships, combining hypergraph message passing with state-space temporal modeling and geometric inductive biases (*IEEE Access*, 2026).
- **Adaptive feature fusion:** Developed a feature-wise temporal-spatial gating mechanism using Kolmogorov–Arnold Networks to learn adaptive feature weighting and nonlinear interactions for representation fusion prior to classification (*Pattern Recognition Letters*, 2026).
- **Equivariant and continual learning:** Developed a patient-adaptive graph learning approach combining equivariant GNNs with continual learning to preserve structured representations under distribution shift (*Neurocomputing*, 2026).
- **Manifold-valued modeling:** Developing graph representations on Log-Euclidean SPD manifolds with persistent homology-based regularisation for dynamic multivariate signal modelling and prediction.

- **Equivariant learning survey:** Authored a survey of symmetry-group equivariant convolutions and representation learning methods for non-Euclidean data (*International Journal of Data Science and Analytics*, 2026).

Teaching Assistant

Jun 2013 – May 2015

Video Processing Lab, Indian Institute of Technology Bombay (IIT-B), Mumbai, India

- Built a person-independent HOG+HOOF pipeline for human activity recognition in far-field, crowded video as part of M.Tech research.
- Contributed to research on automated blood cell counting and wavelet-based beat detection; assisted instruction for three offerings of EE340 (Communications Lab).

INDUSTRY EXPERIENCE

Associate Applications Developer

Jan 2011 – May 2012

Oracle Financial Services Software (OFSS), Bengaluru, India

- Developed and maintained components of the Oracle FLEXCUBE core banking platform using Java, PL/SQL, and Oracle Database.
- Built and deployed enterprise Java (EJB) client applications and managed source code using IBM ClearCase in a collaborative development environment.
- Generated operational and analytical reports using Oracle BI Enterprise Edition (OBIEE).

Systems Engineer

Jun 2010 – Jan 2011

Infosys Technologies Ltd., Mysore, India

- Completed professional training in C, C++, Java, PL/SQL, UNIX, and COBOL, followed by development of business application prototypes.

EDUCATION

Doctor of Philosophy, Avionics

Aug 2022 – Present

Indian Institute of Space Science and Technology (IIST), Thiruvananthapuram, India

Dissertation: Geometric Deep Learning for Seizure Detection and Prediction from EEG Signals

Research focus: Structured and Geometric Representation Learning for Multivariate Neural Signals

Supervisor: Prof. Deepak Mishra | *Expected Completion:* December 2026

Master of Technology, Communication Engineering

2013 – 2015

Indian Institute of Technology Bombay (IIT-B), Mumbai, India

Thesis: Human Activity Recognition in Far Field of View. Designed HOG and HOOF descriptors for person-independent analysis, extending the framework to crowded scenes.

Bachelor of Technology, Electronics and Communication Engineering

2006 – 2010

National Institute of Technology Calicut (NIT-C), Calicut, India

SELECTED PUBLICATIONS

1. R. Basheer and D. Mishra. "Learning Feature-Wise Temporal-Spatial Gating with Kolmogorov-Arnold Networks for EEG Seizure Detection." *Pattern Recognition Letters*, 2026. DOI: 10.1016/j.patrec.2026.07.016
2. R. Basheer and D. Mishra. "Neural State-Space Hypergraph Networks: EEG Seizure Detection via Distributional and Geometric Inductive Biases." *IEEE Access*, 2026. DOI: 10.1109/ACCESS.2026.3702760
3. R. Basheer and D. Mishra. "Harnessing Continual Learning and Equivariant GNNs for Seizure Detection in EEG Time Series." *Neurocomputing*, vol. 663, 2026. DOI: 10.1016/j.neucom.2025.131975
4. R. Basheer and D. Mishra. "Symmetry Group Equivariant Convolutions for Representation Learning: A Survey." *International Journal of Data Science and Analytics*, 22(1), 2026. DOI: 10.1007/s41060-025-00994-7
5. R. Basheer, A. H. Daraie, and D. Mishra. "Hypercube-S4GNN: A Multi-Edge Graph Approach Using State Space Models on Multivariate EEG for Seizure Detection." *IEEE International Workshop on Machine Learning for Signal Processing (MLSP)*, 2024. DOI: 10.1109/MLSP58920.2024.10734832

6. C. P. Akhila and R. Basheer. “RLS Classification of Hyperspectral Image after Fusion and Recursive Filtering.” *International Conference on Information Science*, 2016. DOI: 10.1109/INFOSCI.2016.7845317

TEACHING & SERVICE

Teaching Experience

Aug 2015 – Present

Presidency University, Bengaluru (2018–Present, on sabbatical since 2022);

Acharya Institute of Technology, Bengaluru (2017–2018);

MES College of Engineering, Kuttippuram (2015–2016)

- Taught undergraduate and postgraduate electronics courses and supervised student projects in hyperspectral image classification and satellite-image analysis.

Reviewer: IEEE Access, Neurocomputing.

Awards: Senior Research Fellowship (2024–Present) and Junior Research Fellowship (2022–2024), IIST; Teaching Assistant Fellowship, IIT-B (2013–2015).